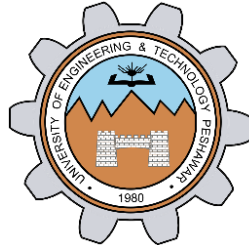


Keypad interfacing (port multiplexing)

LAB # 09



Spring 2024

CSE-307L Microprocessor Based System Design Lab

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Registration No.: **21PWCSE2059**

Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Dr. Asif Ali Khan

Date:

12th May 2024

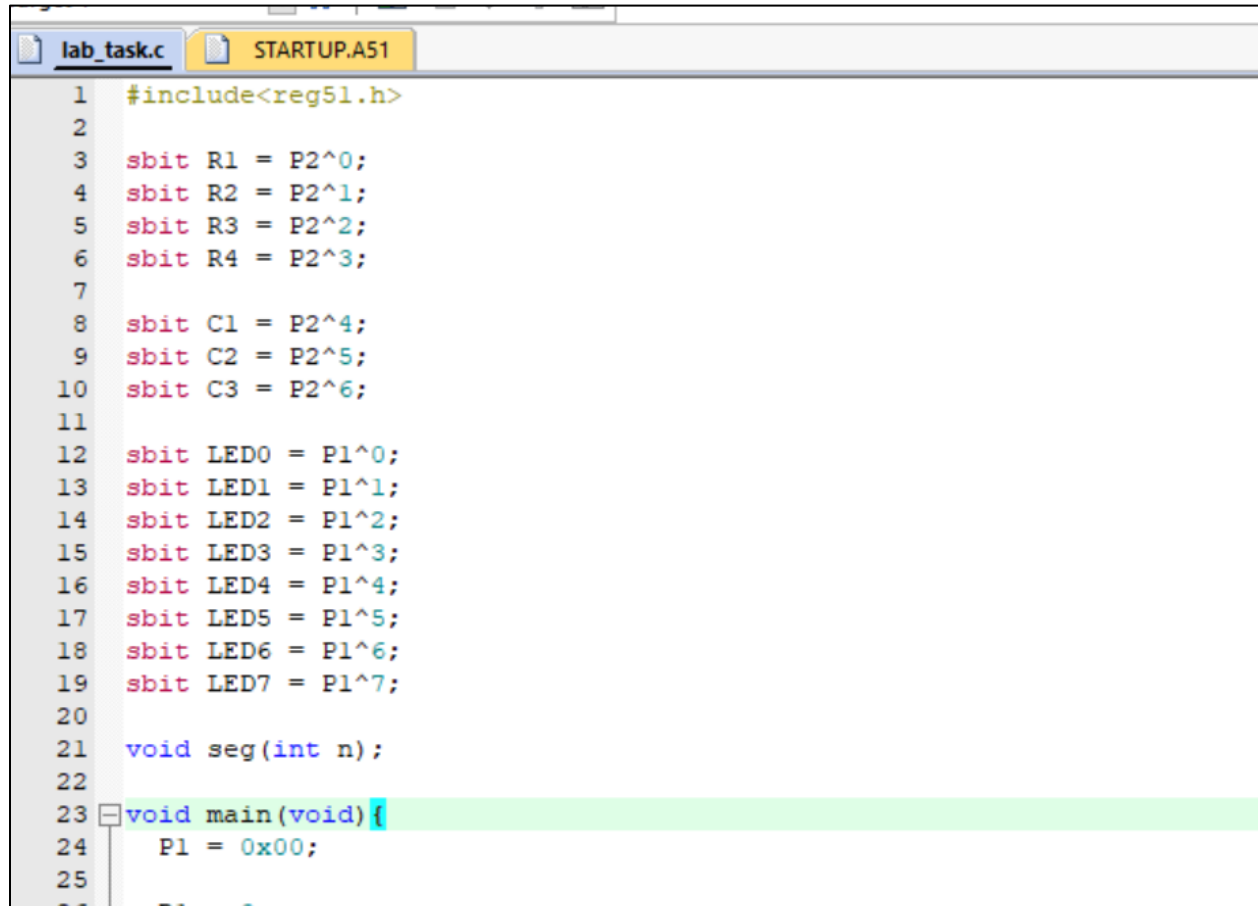
Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Tasks:

Run the program given in the lecture

Display Numbers from 1 to 9 on the seven segment display and ON the corresponding LED's attached with P1 Note: Attach the seven segment with P3 and keypad with P1

Code:



```
1  #include<reg51.h>
2
3  sbit R1 = P2^0;
4  sbit R2 = P2^1;
5  sbit R3 = P2^2;
6  sbit R4 = P2^3;
7
8  sbit C1 = P2^4;
9  sbit C2 = P2^5;
10 sbit C3 = P2^6;
11
12 sbit LED0 = P1^0;
13 sbit LED1 = P1^1;
14 sbit LED2 = P1^2;
15 sbit LED3 = P1^3;
16 sbit LED4 = P1^4;
17 sbit LED5 = P1^5;
18 sbit LED6 = P1^6;
19 sbit LED7 = P1^7;
20
21 void seg(int n);
22
23 void main(void) {
24     P1 = 0x00;
25 }
```

```
lab_task.c | STARTUP.A51
22
23 void main(void) {
24     P1 = 0x00;
25
26     R1 = 0;
27     R2 = R3 = R4 = 1;
28
29     if(C1 == 0 && C2 == 1 && C3 == 1) { //1 Pressed
30         LED0 = 1;
31         seg(1);
32     }
33
34     if(C1 == 1 && C2 == 0 && C3 == 1) { //2 Pressed
35         LED1 = 1;
36         seg(2);
37     }
38
39     if(C1 == 1 && C2 == 1 && C3 == 0) { //3 Pressed
40         LED2 = 1;
41         seg(3);
42     }
43
44
45     R2 = 0;
46     R1 = R3 = R4 = 1;
```

```
lab_task.c | STARTUP.A51
46     R1 = R3 = R4 = 1;
47     if(C1 == 0 && C2 == 1 && C3 == 1) { //4 Pressed
48         LED3 = 1;
49         seg(4);
50     }
51
52     if(C1 == 1 && C2 == 0 && C3 == 1) { //5 Pressed
53         LED4 = 1;
54         seg(5);
55     }
56
57     if(C1 == 1 && C2 == 1 && C3 == 0) { //6 Pressed
58         LED5 = 1;
59         seg(6);
60     }
61
62
63     R3 = 0;
64     R1 = R2 = R4 = 1;
65
66     if(C1 == 0 && C2 == 1 && C3 == 1) { //7 Pressed
67         LED6 = 1;
68         seg(7);
69     }
70
```

```
lab_task.c  STARTUP.A51
70
71 if(C1 == 1 && C2 == 0 && C3 == 1){//8 Pressed
72     LED7 = 1;
73     seg(8);
74 }
75
76 if(C1 == 1 && C2 == 1 && C3 == 0){//9 Pressed
77
78     seg(9);
79 }
80
81 R4 = 0;
82 R1 = R2 = R3 = 1;
83
84 // if(C1 == 0 && C2 == 1 && C3 == 1){//* Pressed
85 //     seg(10);
86 // }
87
88 if(C1 == 1 && C2 == 0 && C3 == 1){//0 Pressed
89     seg(0);
90 }
91
92 // if(C1 == 1 && C2 == 1 && C3 == 0){//# Pressed
93 //     seg(11);
94 // }
```

```
lab_task.c  STARTUP.A51
94 // }
95 }
96
97
98 void seg(int n){
99
100     if(n == 0)
101         P3 = 0x40; //0
102
103     else if(n == 1)
104         P3 = 0x79; //1
105
106     else if(n == 2)
107         P3 = 0x24; //2
108
109     else if(n == 3)
110         P3 = 0x30; //3
111
112     else if(n == 4)
113         P3 = 0x19; //4
114
115     else if(n == 5)
116         P3 = 0x12; //5
117
118     else if(n == 6)
```

```

115     else if(n == 5)
116         P3 = 0x12; //5
117
118     else if(n == 6)
119         P3 = 0x02; //6
120
121     else if(n == 7)
122         P3 = 0x78; //7
123
124     else if(n == 8)
125         P3 = 0x00; //8
126
127     else if(n == 9)
128         P3 = 0x18; //9
129 }

```

Proteus Simulation:

